



Write $M[x := N, y := L]$ for the *simultaneous* substitution: every free x and every free y in M is replaced at the same time, by N and by L respectively. On the right below stands the *sequential* substitution of the lecture, one variable after the other.

$$M[x := N, y := L] \equiv M[x := N][y := L]$$

Show that this is not always true, and state a condition on y and N under which the two sides do agree.

Nederpelt & Geuvers, *Type Theory and Formal Proof* (CUP, 2014), Exercise 1.6.

I Definitions

Substitution (Def. 1.6.1). With $\equiv$ syntactic identity,

$$\begin{aligned} x[x := N] &\equiv N, & y[x := N] &\equiv y \quad (y \neq x), \\ (PQ)[x := N] &\equiv (P[x := N])(Q[x := N]), & (\lambda y. P)[x := N] &\equiv \lambda z. P^{y \rightarrow z}[x := N], \end{aligned}$$

where in the last clause $\lambda z. P^{y \rightarrow z}$ is an α -variant of $\lambda y. P$ with $z \notin \text{FV}(N)$, and $P^{y \rightarrow z}$ is P with every free y renamed to z .

Free variables (Def. 1.4.1). $\text{FV}(x) = \{x\}$, $\text{FV}(PQ) = \text{FV}(P) \cup \text{FV}(Q)$, $\text{FV}(\lambda x. P) = \text{FV}(P) \setminus \{x\}$.

Name _____

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Answer

Out of room? Continue on the back of the page.